Enhanced Drug-Target Interaction Prediction Through Hybrid Deep Learning Approaches

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Abstract

Accurately predicting drug–target interactions (DTIs) is a cornerstone of modern drug discovery, enabling efficient identification of therapeutic candidates, drug repurposing opportunities, and potential off-target effects. Despite recent advances in deep learning, existing architectures face persistent challenges: weak cross-modal feature integration due to shallow fusion of heterogeneous drug and protein representations, poor generalisation in cold-start scenarios, and limited interpretability.

This thesis presents **H-GNN-RNN-DTI**, a hybrid deep learning model integrating a 3-layer Hierarchical Graph Attention Network (H-GAT) for drug encoding with a 2-layer Bidirectional Gated Recurrent Unit (BiGRU) for protein encoding, fused through a novel Gated Cross-Attention (GCA) mechanism. Three research objectives were addressed: (1) architectural design and evaluation; (2) cold-start enhancement through imbalance-aware training, larger protein language models, and large-scale pre-training; and (3) attention-based interpretability with biochemical validation.

Key results: Gated Cross-Attention outperformed concatenation fusion on 3/8 DTI configurations ($p<0.05$) and 5/6 Davis DTA metrics ($p<0.05$). The 1:5 negative-sampling ratio improved Yamanishi_e cold_target AUROC from 0.438 to 0.706 ($+0.268$, $p<0.01$). ESM-2 150M improved Luo_17 cold_target by $+0.069$ AUROC ($p<0.01$). BindingDB pre-training improved Luo_17 cold_target by $+0.055$ AUROC ($p<0.01$). Drug attribution faithfulness advantage was $+0.106$ (Ensemble); DHFR binding-site AUROC was 0.613 with enrichment factor $2.81\times$ at 10%.

Keywords: Drug-target interaction, Graph attention network, BiGRU, Gated cross-attention, Cold-start generalisation, Interpretability, Pharmacophore, Protein language model.

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CHAPTER 1

Introduction

1.1 Background and Motivation

Identifying the interactions between drugs and their target proteins is a fundamental step in the drug discovery pipeline. Beyond understanding disease mechanisms, accurate DTI prediction enables drug repurposing and anticipation of adverse side effects (27; 33). The traditional drug development process is notoriously slow and expensive – approximately $2.6 billion and 15 years per approved drug (11) – creating an economic bottleneck that computational approaches aim to alleviate.

Deep learning has emerged as a powerful paradigm for DTI prediction, enabling automatic feature extraction from raw molecular representations without hand-crafted features (29). Two modalities are central: the graph structure of drug molecules and the sequential nature of protein amino acid sequences. Graph Neural Networks (GNNs) capture molecular topology; Recurrent Neural Networks (RNNs) capture protein sequence context. The dominant trend is to combine both modalities in hybrid architectures (22; 24).

1.2 Problem Statement

Despite significant advances, three critical limitations persist:

1. **Sub-optimal cross-modal feature integration**: Models such as GSRF-DTI (37) employ static concatenation fusion preventing meaningful drug–protein interaction at the feature level. DTKGIN (17) relies on inner-product scoring without dynamic attention-based fusion.
2. **Inadequate cold-start generalisation**: Most models memorise training-distribution patterns rather than learning transferable biochemical principles. Pre-trained protein language model embeddings (ESM-2, ProtT5) that encode evolutionary knowledge are not leveraged by current methods.
3. **Limited interpretability**: High-performing models lack atom-level or residue-level attribution mechanisms, constraining their utility for rational drug design and structure–activity relationship studies.

1.3 Research Questions

RQ1 How can a Gated Cross-Attention fusion mechanism effectively integrate GNN-based drug representations with RNN-based protein representations, and does this approach improve DTI prediction compared to static concatenation?

RQ2 To what extent can pre-trained molecular and protein language embeddings, combined with imbalance-aware learning strategies, improve generalisation to unseen drugs, unseen targets, and cross-dataset scenarios?

RQ3 Can an attention-enhanced hybrid model provide interpretable and biologically meaningful insights by identifying key binding residues and pharmacophoric features?

1.4 Research Objectives

1. To design and evaluate a hybrid model integrating structural graph and sequence encoders through a Gated Cross-Attention mechanism, achieving statistically significant performance improvements over concatenation-based fusion.
2. To develop and evaluate enhancement strategies – imbalance-aware training, larger protein language models, and BindingDB pre-training – with the aim of improving performance on cold-start scenarios.
3. To implement and benchmark an attention-based interpretability layer validated against PDB binding-site annotations and pharmacophore standards.

1.5 Scope of the Research

The research addressed computational DTI/DTA prediction using public benchmark datasets. Drug molecules were represented as molecular graphs (SMILES $\rightarrow$ PyG Data objects via RDKit, with ChemBERTa embeddings). Proteins were represented using ESM-2 embeddings (8M and 150M parameter variants). Datasets: Yamanishi_e and Luo_17 (DTI binary classification); DAVIS and KIBA (DTA regression); BindingDB (pre-training, 49,499 pairs). The research excluded wet-lab validation, non-small-molecule ligands, and large-scale industrial deployment.

1.6 Significance of the Research

This research makes four primary contributions: (1) a principled cross-modal fusion mechanism enabling context-aware drug–protein feature integration; (2) a systematic study of cold-start enhancement strategies revealing a non-additive hierarchy of improvements; (3) demonstration that attention-based attributions recover binding sites without structural supervision; and (4) a

fully inductive, dual-task architecture achieving cold_target AUROC 0.778 on Luo_17 versus 0.362 for the transductive GSRF-DTI baseline.

1.7 Organisation of the Thesis

Chapter 2 reviews computational DTI methodologies and related work. Chapter 3 presents the H-GNN-RNN-DTI architecture, datasets, preprocessing, training strategy, and evaluation protocol. Chapter 4 presents and discusses results for all three objectives. Chapter 5 concludes with contributions, limitations, and future directions.

CHAPTER 2

Literature Review

2.1 Introduction

This chapter reviews the evolution of computational methodologies for DTI prediction, from early physics-based simulations to modern hybrid deep learning architectures, identifying persistent limitations that motivated this work.

2.2 Computational Methodologies for DTI Prediction

2.2.1 Molecular Docking-Based Approaches

Molecular docking simulates the physical binding of a ligand to the 3D structure of a target protein, providing detailed structural insights (5). These methods are computationally expensive, depend on high-resolution 3D protein structures unavailable for many targets, and are impractical for large-scale virtual screening (16; 7).

2.2.2 Ligand-Based Methods

Ligand-based methods predict activity based on structural similarity to known active ligands. Their effectiveness diminishes for targets with few known binders and for novel chemotypes without structural analogues (12). These limitations drove the field toward data-driven machine learning approaches.

2.2.3 Traditional Machine Learning and Network-Based Methods

Classical ML methods – Gradient Boosting, Random Forest, SVM, MLP – were applied to DTI prediction using hand-crafted molecular fingerprints. Network-based methods exploited heterogeneous biological network topology (drug–drug, protein–protein, DTI networks) via algorithms such as Random Walk with Restart (4; 30). Persistent challenges include absence of high-quality negative samples, class imbalance, and inability to generalise to novel entities.

2.2.4 Deep Learning in DTI Prediction

Deep learning enabled automatic feature learning from raw molecular representations (29; 14). CNNs, RNNs, and GNNs achieved strong performance on benchmark datasets. However,

single-architecture models remain limited: GNNs miss sequential protein context; RNNs miss molecular topology. Cold-start scenarios, data sparsity, and interpretability remain critical challenges (22; 9).

2.3 Foundational Deep Learning Architectures

2.3.1 Recurrent Neural Networks

RNNs maintain a hidden state over sequences, capturing temporal dependencies (1; 19). LSTM and GRU variants address long-range dependency learning. Bidirectional variants (BiGRU, BiLSTM) process sequences in both directions. A known limitation is the inability to natively model non-linear molecular topology.

2.3.2 Graph Neural Networks

GNNs operate on graph-structured data through iterative neighbourhood aggregation (message passing) (35). Graph Attention Networks (GAT) weight neighbour contributions via learned attention coefficients. GraphDTA (20) demonstrated that GNN-based DTA models outperform CNN-based models. Oversmoothing – repeated message passing leading to uniform node representations – is addressed by residual connections (8).

2.4 Rationale for Hybrid Deep Learning Models

The "feature bottleneck" of single-architecture models motivates hybrid designs. A dual-branch system assigns a specialised encoder to each modality: GNN for the drug molecular graph, BiGRU for the protein sequence. Prior work combining CNNs with LSTMs (e.g., DeepGS **(author?)** 36) confirmed that multi-modal integration yields more comprehensive representations.

2.5 Critical Analysis of Related Methods

2.5.1 GSRF-DTI (37)

GSRF-DTI integrates DeepWalk, GraphSAGE, and Random Forest, achieving AUROC=0.9818 under random 5-fold CV with the full DrugBank knowledge graph. Key limitations: (1) static concatenation prevents cross-modal interaction; (2) transductive design fails on cold_target (AUROC=0.362 in our re-implementation); (3) no atom-level attribution; (4) authors acknowledge inadequate handling of data imbalance.

2.5.2 TTModel (13)

TTModel uses knowledge graph embedding with textual and entity-type features, achieving AUPR=0.95 on GPCR. Limitations: no molecular graph encoding; no direct protein sequence

processing; no cross-modal attention fusion; no atom- or residue-level attribution.

2.5.3 DTKGIN (17)

DTKGIN introduces intent nodes within a GNN paradigm, achieving AUC=0.894 on Yamanishi_08 cold-drug splits. Limitations: no dedicated molecular graph GNN or sequence RNN encoder; inner-product scoring; no pre-trained embeddings; high variance on cold_drug ($\pm$0.100 AUROC).

2.6 Current Research Gaps

Cross-modal integration Static concatenation prevents the drug representation from informing protein representation. Dynamic cross-attention – where each drug atom attends to protein residues – enables context-dependent enrichment (25; 23).

Cold-start generalisation Most models fail when evaluated on unseen drugs (cold_drug) or unseen proteins (cold_target). Pre-trained foundation models (ChemBERTa, ESM-2) encode transferable biochemical knowledge applicable to novel entities (12; 26).

Interpretability Attention-based attribution – which drug atoms and protein residues drive the prediction – provides intrinsic interpretability validated by faithfulness metrics and PDB binding-site comparisons (32; 34).

2.7 Related Work

Recent hybrid models advanced cross-modal fusion and interpretability. iNGNN-DTI (25) employs nested GNN with cross-attention modules. EmbedDTI (10) uses sequence embeddings with GCN and attention for interpretable prediction. HMSA-DTI (3) introduces hierarchical multimodal self-attention. Top-DTI (26) integrates topological deep learning with large language model embeddings for cold-start robustness. MutualDTA (34) introduces explicit drug atom to protein residue attention for interpretable affinity prediction.

2.8 Summary

The literature confirms that hybrid deep learning has advanced DTI prediction, but persistent gaps remain in dynamic fusion, cold-start generalisation, and interpretability. H-GNN-RNN-DTI addresses all four gaps simultaneously through its architectural design and enhancement strategies.

CHAPTER 3

Methodology

3.1 Research Design

This study adopted a quantitative, computational design structured around three empirical objectives: architecture design and evaluation, enhancement strategy benchmarking, and interpretability validation. The research followed an experimental approach involving model development, training and evaluation on public DTI/DTA benchmarks, baseline comparison, and interpretability case studies with PDB-validated binding sites.

3.2 Datasets

Table 3.1: Datasets used in this research.

Dataset	Task	Drugs	Proteins	Pos. Pairs	Use
Yamanishi_e	DTI (binary)	68	26	445	Training/evaluation
Luo_17	DTI (binary)	708	1,512	1,923	Training/evaluation
DAVIS	DTA (pKd)	68	442	30,056	Affinity regression
KIBA	DTA (score)	229	229	118,254	Affinity regression
BindingDB	DTI (binary)	47,079	diverse	49,499	Pre-training

Yamanishi_e is a gold-standard small enzyme benchmark (31). Luo_17 is a larger heterogeneous benchmark (18). DAVIS and KIBA are standard DTA benchmarks (21). BindingDB was used as a large-scale pre-training source (49,499 balanced pairs extracted from 47,079 unique SMILES).

3.2.1 Data Splitting Protocol

Four splitting strategies were implemented:

1. **Random split** (warm-start): 70/10/20 train/val/test.
2. **Cold_drug split**: Test drugs never appear in training.
3. **Cold_target split**: Test proteins never appear in training.
4. **Cold_pair split**: Test drug–protein pairs not observed in training.

3.3 Data Preprocessing

3.3.1 Drug Representation

Drug molecules were processed from SMILES strings into PyG `Data` objects using RDKit. Atom node features included: atomic number, degree, hybridisation, aromaticity, formal charge, hydrogen count, and ring membership. Bond edge features included: bond type, conjugation, and ring membership. Pre-trained ChemBERTa embeddings (`seyonec/ChemBERTa-zinc-base-v1`, 768-dim CLS token) were fused with atom features inside the DrugEncoder.

3.3.2 Protein Representation

Protein sequences were encoded using ESM-2 protein language models. For Yamanishi_e (per-residue BiGRU path): ESM-2 8M (`facebook/esm2_t6_8M_UR50D`, 320-dim, shape $B \times L \times 320$). For Luo_17 (mean-pooled linear path): ESM-2 8M (shape $B \times 320$) or ESM-2 150M (`facebook/esm2_t12_35M_UR50D`, 480-dim, shape $B \times 480$). A learned linear projection ($480 \rightarrow 320$) was applied inside the model when 150M embeddings were used.

3.3.3 Label Processing

For DTI binary classification, confirmed interactions were labelled positive (1). Negative pairs were sampled from non-interacting combinations at ratios of 1:1, 1:5, or 1:10 (positive:negative). For DTA regression, binding affinity values were used directly (pKd for DAVIS, KIBA scores for KIBA).

3.4 H-GNN-RNN-DTI Architecture

The model consists of: (1) DrugEncoder (Hierarchical GAT), (2) ProteinEncoder (BiGRU with attention pooling), (3) GatedCrossAttention fusion, and (4) task-specific prediction heads.

3.4.1 Drug Encoder — Hierarchical Graph Attention Network

Drug molecules are encoded through three stacked `ResidualGATBlock` layers, each with 8 attention heads. The message passing update at layer ℓ for atom i is:

$$\mathbf{h}_i^{(\ell)} = \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(\ell)} \cdot \mathbf{W}^{(\ell)} \mathbf{h}_j^{(\ell-1)}\right) \tag{3.1}$$

where $\mathcal{N}(i)$ are atom i's bonded neighbours, α_{ij} are attention-normalised coefficients, $\mathbf{W}^{(\ell)}$ is a learnable weight matrix, and σ is ELU activation. A global drug embedding is derived by

concatenating mean-pooling and max-pooling, then fusing with ChemBERTa:

$$\mathbf{h}_{\text{drug}} = \text{MLP}(\text{concat}(\text{MeanPool}, \text{MaxPool}) \oplus \mathbf{W}_{\text{cb}}\mathbf{e}_{\text{ChemBERTa}}) \tag{3.2}$$

3.4.2 Protein Encoder — Bidirectional GRU

Per-residue ESM-2 embeddings $\mathbf{x}_t$ are processed by a 2-layer BiGRU:

$$\overrightarrow{\mathbf{h}}_t = \text{GRU}(\mathbf{x}_t, \overrightarrow{\mathbf{h}}_{t-1}), \quad \overleftarrow{\mathbf{h}}_t = \text{GRU}(\mathbf{x}_t, \overleftarrow{\mathbf{h}}_{t+1}) \tag{3.3}$$

$$\mathbf{h}_t = \text{concat}(\overrightarrow{\mathbf{h}}_t, \overleftarrow{\mathbf{h}}_t) \tag{3.4}$$

Attention pooling derives a compact sequence-level representation:

$$e_t = \mathbf{v}^\top \tanh(\mathbf{W}_a\mathbf{h}_t + \mathbf{b}_a), \quad \alpha_t = \frac{\exp(e_t)}{\sum_s \exp(e_s)}, \quad \mathbf{h}_{\text{protein}} = \sum_t \alpha_t \mathbf{h}_t \tag{3.5}$$

The weights $\alpha_t \in [0, 1]$ serve directly as per-residue importance scores for the interpretability module.

3.4.3 Gated Cross-Attention Fusion

Given $\mathbf{h}_{\text{drug}}$ and $\mathbf{h}_{\text{protein}}$, bidirectional scaled dot-product cross-attention is computed:

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}_d\mathbf{K}_p^\top}{\sqrt{d_k}}\right)\mathbf{V}_p \tag{3.6}$$

where $\mathbf{Q}_d = \mathbf{W}_Q\mathbf{h}_{\text{drug}}$, $\mathbf{K}_p = \mathbf{W}_K\mathbf{h}_{\text{protein}}$, $\mathbf{V}_p = \mathbf{W}_V\mathbf{h}_{\text{protein}}$. A gating mechanism controls information flow:

$$\mathbf{g} = \sigma(\mathbf{W}_g \cdot \text{concat}(\mathbf{h}_{\text{drug}}, \mathbf{A})), \quad \mathbf{h}'_{\text{drug}} = \mathbf{g} \odot \mathbf{h}_{\text{drug}} + (1 - \mathbf{g}) \odot \mathbf{A} \tag{3.7}$$

The same gating is applied in the reverse direction (protein attending to drug). The fused representation is:

$$\mathbf{h}_{\text{fused}} = \text{MLP}\big(\text{concat}(\mathbf{h}'_{\text{drug}}, \mathbf{h}'_{\text{protein}})\big) \tag{3.8}$$

3.4.4 Prediction Heads

For DTI binary classification, FocalLoss is applied to MLP logits:

$$\mathcal{L}_{\text{focal}} = -\alpha_t \cdot (1 - p_t)^\gamma \cdot \log(p_t), \quad \gamma = 2 \tag{3.9}$$

For DTA regression, MSE loss is used:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2 \tag{3.10}$$

3.5 Training Strategy

Optimiser: Adam (lr=10^{-3}, weight_decay=10^{-4}). **Scheduler:** CosineAnnealingLR ($T_{\max} = 100$, $\eta_{\min} = 10^{-6}$). **Early stopping:** patience=15 on validation AUROC (DTI) or MSE (DTA). **Gradient clipping:** max_norm=1.0. **Batch size:** 64 on CPU.

For BindingDB pre-training (Objective 2 Phase 3): lr=10^{-4}, 50 epochs, batch=64. Best-epoch checkpoint (epoch 42, test AUROC=0.728) was used to initialise encoder and fusion weights; task head was re-initialised.

3.6 Evaluation Protocol

Primary DTI metric: AUROC. Secondary: AUPRC, F1-score.

DTA metrics: MSE (primary), MAE, RMSE, R^2, Pearson r, Spearman ρ.

Statistical testing: Two-tailed paired t-test (df=4, $n = 5$ seeds, $\alpha = 0.05$). All results reported as mean $\pm$ std over 5 random seeds.

3.7 Interpretability Module

Input Gradient: $\text{importance}[i] = \|\partial\text{score}/\partial\mathbf{x}_i\|_2$

Graph Grad-CAM: $\text{cam}[i] = \text{ReLU}\left(\text{mean}_c\left(\frac{\partial\text{score}}{\partial h_{i,c}} \cdot h_{i,c}\right)\right)$

Ensemble: $\text{ens}[i] = 0.5 \times \text{norm}(\text{InputGrad}[i]) + 0.5 \times \text{norm}(\text{GradCAM}[i])$

AttentionPooling: Per-residue weights α_t from Equation 3.5.

Faithfulness AUC: Area under the DTI-probability vs. progressive-masking curve (atoms zeroed in descending attribution order). Advantage = faithfulness_AUC − random_AUC.

Pharmacophore precision@K: Top-$k\%$ attributed atoms matched against SMARTS patterns for HBD, HBA, Arom, Hydro. Enrichment = precision@K / random baseline.

PDB binding-site AUROC: Residue-level AUROC between α_t scores and binary binding labels (residues within 4.5 Å of ligand heavy atoms in crystal structure).

CHAPTER 4

Results and Discussion

4.1 Experimental Setup Summary

All experiments used fold 0, 5 random seeds (seeds 1–5), Adam optimiser (lr=10^{-3}), CosineAnnealingLR, early stopping (patience=15), gradient clipping (1.0), and batch size 64 on CPU. The baseline configuration (C0) used ESM-2 8M embeddings and 1:1 negative sampling.

4.2 Objective 1: H-GNN-RNN-DTI Architecture Design and Evaluation

4.2.1 Fusion Strategy: Cross-Attention vs. Concatenation

Table 4.1 reports AUROC (mean ± std, 5 seeds) for H-GNN-RNN-DTI and ConcatBaseline.

Table 4.1: AUROC – H-GNN-RNN-DTI vs. ConcatBaseline (5 seeds, paired t-test). ✓=significantly better, ×=significantly worse, n.s.=not significant.

Dataset	Split	H-GNN-RNN-DTI	ConcatBasel	Δ	p-value
yamanishi_e	random	**0.756±0.125**	0.559±0.093	+0.197	<0.05 ✓
yamanishi_e	cold_drug	**0.774±0.011**	0.638±0.041	+0.136	<0.01 ✓
yamanishi_e	cold_target	0.438±0.045	0.415±0.123	+0.023	n.s.
yamanishi_e	cold_pair	**0.746±0.054**	0.631±0.021	+0.115	<0.05 ✓
luo_17	random	0.946±0.005	**0.948±0.004**	−0.002	n.s.
luo_17	cold_drug	**0.913±0.009**	0.907±0.006	+0.006	n.s.
luo_17	cold_target	0.778±0.014	**0.813±0.013**	−0.035	<0.01 ×
luo_17	cold_pair	**0.948±0.002**	0.943±0.005	+0.005	n.s.

H-GNN-RNN-DTI is significantly better on 3/8 configurations (yamanishi_e random, cold_drug, cold_pair; all p<0.05). The largest advantage is on yamanishi_e cold_drug (t=6.55, p<0.01, Δ=+0.136 AUROC). ConcatBaseline is significantly better on luo_17 cold_target (p<0.01, Δ=−0.035). The pattern is consistent: Gated Cross-Attention provides the largest benefit on small, sparse datasets and cold-start splits; on large datasets, concatenation fusion is competitive.

4.2.2 Encoder Ablation: GNN-Only and RNN-Only

Table 4.2: Encoder ablation – AUROC mean (5 seeds). Bold = best per row.

Dataset	Split	H-GNN	Concat	GNN-Only	RNN-Only
yamanishi_e	random	**0.756**	0.559	0.527	0.674
yamanishi_e	cold_drug	**0.774**	0.638	0.554	0.625
yamanishi_e	cold_target	0.438	0.415	0.508	0.412
yamanishi_e	cold_pair	**0.746**	0.631	0.515	0.704
luo_17	random	**0.946**	0.948	0.902	0.875
luo_17	cold_drug	**0.913**	0.907	0.707	0.909
luo_17	cold_target	0.778	0.813	**0.901**	0.603
luo_17	cold_pair	**0.948**	0.943	0.901	0.888

The ablation reveals symmetric encoder complementarity: the drug GNN drives cold_drug generalisation (GNN-Only collapses on cold_drug: luo_17 -0.206, p$<$0.001); the protein BiGRU drives cold_target generalisation (RNN-Only collapses on cold_target: luo_17 -0.174, p$<$0.001). The anomalous GNN-Only result on luo_17 cold_target (0.901) reflects drug promiscuity: certain scaffolds bind broadly regardless of protein identity, and incorporating noisy unseen-protein ESM-2 embeddings via cross-attention hurts performance. The full model balances both modalities and leads on 6/8 configurations overall.

4.2.3 DTA Regression: DAVIS and KIBA

Table 4.3: DTA regression – mean $\pm$ std (5 seeds). Bold = sig. better (p$<$0.05).

Dataset	Model	MSE↓	MAE↓	R^2 ↑	Pearson r ↑	Spearman ρ ↑
Davis	H-GNN-RNN-DTI	**0.379±0.01**	**0.415±0.0**	**0.508±0.0**	**0.721±0.0**	**0.610±0.011**
Davis	ConcatBaseline	0.468±0.029	0.444±0.02	0.391±0.03	0.626±0.030	0.565±0.019
KIBA	H-GNN-RNN-DTI	**0.274±0.01**	0.354±0.0	**0.613±0.0**	**0.787±0.0**	**0.759±0.016**
KIBA	ConcatBaseline	0.304±0.011	**0.351±0.00**	0.570±0.01	0.756±0.010	0.747±0.011

H-GNN-RNN-DTI outperforms ConcatBaseline on DTA regression. On DAVIS: all 6 metrics improved (MSE $\Delta = -0.090$, p=0.003; Pearson r $\Delta = +0.095$, p=0.003). On KIBA: primary metrics improved significantly (MSE $\Delta = -0.031$, p=0.009; Pearson $\Delta = +0.032$, p=0.006; R^2 $\Delta = +0.043$, p=0.009). The Gated Cross-Attention advantage extends to continuous affinity prediction, confirming task-agnostic representational quality.

Table 4.4: AUROC on Luo_17 – all models. ‡=re-implementation with 3 seeds. ⌗=near-random (transductive baseline fails cold_target).

Model	random	cold_drug	cold_target	cold_pair
H-GNN-RNN-DTI	**0.946±0.005**	**0.913±0.009**	0.778±0.014	**0.948±0.002**
ConcatBaseline	0.948±0.004	0.907±0.006	**0.813±0.013**	0.943±0.005
GSRF-DTI‡	0.767±0.053	0.775±0.047	0.362±0.055	0.859±0.016

Table 4.5: AUROC on Yamanishi_e – all models.

Model	random	cold_drug	cold_target	cold_pair
H-GNN-RNN-DTI	0.756±0.125	**0.774±0.011**	0.438±0.045	0.746±0.054
ConcatBaseline	0.559±0.093	0.638±0.041	0.415±0.123	0.631±0.021
DTKGIN‡	**0.869±0.011**	0.747±0.100	**0.557±0.082**	**0.863±0.015**

4.2.4 External Baseline Comparison

H-GNN-RNN-DTI outperforms GSRF-DTI on all four Luo_17 splits. The key advantage is cold_target: 0.778 vs. 0.362 – GSRF-DTI is transductive and cannot embed proteins absent from the PPI network. H-GNN-RNN-DTI processes each protein from its sequence embedding independently, enabling cold-target prediction. On Yamanishi_e, DTKGIN outperforms H-GNN-RNN-DTI on random, cold_target, and cold_pair, attributable to its use of drug chemical similarity edges on this small dataset (68 drugs). H-GNN-RNN-DTI leads on cold_drug (0.774 vs. 0.747) with substantially lower variance (±0.011 vs. ±0.100).

4.2.5 Cross-Dataset Generalisation

H-GNN-RNN-DTI trained on Luo_17 achieved AUROC of 0.508 ± 0.020 when evaluated on all of Yamanishi_e (zero drug-protein overlap). This is indistinguishable from random (0.5), confirming the model is discriminative rather than generative: without shared molecular scaffolds or protein families, no zero-shot transfer is possible. This motivates future domain-adaptive pre-training.

4.3 Objective 2: Enhancement Strategies for Cold-Start Generalisation

4.3.1 Phase 2: Imbalance-Aware Training

The 1:5 negative-sampling ratio is the single most impactful enhancement across all Objective 2 experiments. On Yamanishi_e, cold_target improves from 0.438 to 0.706 (+0.268, $p<0.01$) – the largest single gain in the entire study. On Luo_17, cold_target improves from 0.778 to 0.856 (+0.078, $p<0.01$). In contrast, all loss function variants (WeightedBCE, Focal $\gamma \in \{1, 2, 3, 5\}$)

Table 4.6: Phase 2 AUROC – ratio and loss sweep (5 seeds, yamanishi_e). Statistically significant improvements vs. C0 annotated (✓✓=p<0.01; ✓=p<0.05).

Config	random	cold_drug	cold_target	cold_pair
C0 (1:1, Focal)	0.756±0.12	0.774±0.01	0.438±0.04	0.746±0.054
P2-F (1:5)	0.852±0.006	0.816±0.016 ✓✓	**0.706±0.104** ✓✓	0.819±0.019
P2-G (1:10)	0.867±0.03	**0.904±0.01** ✓✓	0.735±0.02 ✓✓	**0.882±0.006** ✓✓
WeightedBCE 1:1	0.744±0.122	0.757±0.023	0.402±0.037	0.772±0.016

produce no statistically significant changes at fixed 1:1 ratio. The gain originates entirely from the number of negative examples, not the loss weighting scheme.

An important trade-off: the 1:5 ratio significantly degrades AUPRC. At 1:10 on Luo_17, cold_target AUPRC drops from 0.702 to 0.330 (-0.372, $p<0.01$). The 1:5 ratio provides the best AUROC improvement with lower AUPRC cost and is the recommended configuration for recall-oriented applications.

4.3.2 Phase 1: Protein Embedding Scale

ESM-2 150M significantly improves Luo_17 cold_target AUROC by $+0.069$ (0.778→0.847, $p<0.01$) and AUPRC by $+0.096$ ($p<0.01$). All Yamanishi_e comparisons are non-significant (all n.s., paired t-test, n=5). The divergence is explained by dataset scale: Luo_17 (1,512 proteins) provides sufficient data for richer 480-dim representations to improve unseen-protein coverage; Yamanishi_e (89 proteins) is too small for the PLM scaling signal to exceed seed variance (std up to 0.125 AUROC). ESM-2 150M is recommended for large-dataset cold_target evaluation but unnecessary for small benchmarks.

4.3.3 Phase 3: BindingDB Pre-Training

BindingDB pre-training significantly improves Luo_17 cold_target AUROC ($+0.055$, $p<0.01$) and AUPRC ($+0.059$, $p<0.01$), but significantly hurts Yamanishi_e cold_target (-0.091, $p<0.05$) and cold_pair (-0.127, $p<0.05$). Two mechanisms explain the asymmetry: (1) *domain alignment* – BindingDB's pharmacologically diverse drug space is similar to Luo_17 but misaligned with Yamanishi_e's narrow enzyme-substrate sub-space; (2) *protein encoder path mismatch* – BindingDB pre-training exercises the linear path of ProteinEncoder (mean-pooled), leaving the BiGRU untouched; fine-tuning on Yamanishi_e uses the BiGRU path, creating conflicting optimisation signals from a pre-biased drug encoder and a random-init BiGRU on only 726 training pairs.

4.3.4 Phase 5: Combined Ablation C0→C3

Table 4.7: Phase 5 ablation AUROC (5 seeds). C3 is the AUROC winner overall.

Config	random	cold_drug	cold_targ	cold_pair
yamanishi_e				
C0 (8M/1:1/none)	0.756	0.774	0.438	0.746
C1 (150M/1:1/none)	0.830	0.743	0.378	0.778
C2 (150M/1:5/none)	0.730	0.773	0.528	0.719
C3 (150M/1:5/BDB)	**0.850**	**0.849**	**0.575**	**0.804**
luo_17				
C0 (8M/1:1/none)	0.946	0.913	0.778	0.948
C1 (150M/1:1/none)	0.952	0.905	0.847	0.952
C2 (150M/1:5/none)	**0.965**	0.851	0.838	0.938
C3 (150M/1:5/BDB)	0.963	0.874	**0.847**	**0.954**

Improvements are *not additive*. C2 (150M + 1:5) underperforms both components alone on key metrics: luo_17 cold_drug degrades by -0.062 vs. C0 ($p<0.01$); yamanishi_e cold_target (0.528 $\pm$ 0.218, n.s.) is far below Phase 2 alone (0.706, $p<0.01$). C3 (adding BindingDB pre-training) partially resolves these conflicts: it achieves the best AUROC on all yamanishi_e splits (cold_drug $+0.075$ vs. C0, $p<0.01$) and ties C1 on luo_17 cold_target ($+0.069$, $p<0.01$). The AUPRC cost from the 1:5 ratio persists through C3.

Hierarchy: negative-sampling ratio > BindingDB pre-training > embedding scale for AUROC on cold-start splits. Embedding scale alone is safest when both AUROC and AUPRC must be maintained.

4.4 Objective 3: Attention-Based Interpretability

4.4.1 Drug Attribution – Faithfulness and Pharmacophore

Pilot case: L-Tryptophan (DB00150) vs. ADORA3 receptor (P19838), Luo_17 C3 checkpoint, DTI prediction probability = 0.869.

The Ensemble method achieves the highest faithfulness advantage ($+0.106$ vs. $+0.020$ random), confirming that Input Gradient and Graph Grad-CAM provide complementary signals (Spearman $\rho = -0.138$ between them). Masking the top-1 atom drops probability from 0.869 to 0.269 (Δ=0.600) vs. only 0.160 for random single-atom masking. Top-5 attributed atoms: indole C3

Table 4.8: Drug attribution faithfulness – DB00150/P19838 (C3 checkpoint).

Method	Faith. AUC	Random AUC	Advantage	PharmAP
Random	+0.169	+0.149	+0.020	0.583
Input Gradient	+0.226	+0.149	+0.077	0.327
Graph Grad-CAM	+0.157	+0.149	+0.008	0.458
Ensemble	**+0.255**	+0.149	**+0.106**	0.423

(1.000), indole NH (0.804), alpha-amino N14 (0.755), ring carbons (0.667, 0.659) – consistent with known tryptophan pharmacology and GPCR binding mechanisms. Pharmacophore precision at k=10%: HBD=0.50 (3.8× random), HBA=0.67 at k=20% (2.5× random).

4.4.2 Protein Attribution – PDB Case Studies

Table 4.9: PDB binding-site validation – residue attribution vs. 4.5 Å annotations.

Case	Protein	DTI Prob.	AUROC	P@10%	EF@10%
DHFR+MTX	P00374 (187 aa)	0.328	**0.613**	**0.316**	**2.81**×
ADA+ADO	P00813 (363 aa)	0.220–0.378	0.327	–	–
Random baseline	–	–	0.528	0.112	1.0×

For DHFR (Methotrexate vs. Dihydrofolate Reductase, PDB 1U72), AttentionPooling achieves AUROC=0.613 and P@10%=0.316 (2.81× random enrichment). Top-attributed residues (Phe31, Ile60–Pro61, Asn64 region) are confirmed by PDB to lie within 4.5 Å of the methotrexate pteridine ring, consistent with known folate-binding pocket contacts. The model identifies binding residues from sequence context alone, without 3D structural training.

For ADA (Adenosine vs. Adenosine Deaminase), low prediction confidence (prob=0.220–0.378) produces residue AUROC=0.327 (below random) – an informative negative confirming that attribution is prediction-conditional and mechanistically coherent.

4.4.3 Attribution Method Comparison

AttentionPooling is the primary protein attribution signal (AUROC=0.613). Protein GradCAM yields AUROC=0.500 (random level): BiGRU hidden-state gradients are not spatially localised to binding residues because the BiGRU learns distributed sequence representations. The ensemble therefore collapses to AttentionPooling for protein attribution. For drug attribution, the Ensemble remains the recommended method.

Table 4.10: Protein attribution method comparison – D00142/P00374 (DHFR).

Method	AUROC	P@10%	Comment
Random	0.528	0.211	Baseline
AttnPool	**0.613**	**0.316**	BiGRU head recovers binding residues
Protein GradCAM	0.500	0.000	BiGRU gradients not spatially localised
Ensemble	0.613	0.316	Collapses to AttnPool only

CHAPTER 5

Conclusion

5.1 Summary of Contributions

This thesis presented H-GNN-RNN-DTI, a hybrid deep learning model for DTI/DTA prediction integrating Hierarchical GAT, BiGRU, and Gated Cross-Attention. Three research objectives were addressed.

Objective 1 (Architecture): Gated Cross-Attention significantly outperformed concatenation fusion on 3/8 DTI configurations ($p<0.05$) and on 5/6 Davis DTA metrics ($p<0.05$). Encoder ablation revealed symmetric complementarity: drug GNN drives cold_drug, protein BiGRU drives cold_target generalisation. Compared to published baselines, the fully inductive design provides 2.15× cold_target AUROC advantage over GSRF-DTI (0.778 vs. 0.362).

Objective 2 (Enhancement): Negative-sampling ratio is the dominant lever – the 1:5 ratio raised yamanishi_e cold_target AUROC from 0.438 to 0.706 ($+0.268$, $p<0.01$), the largest improvement in the study. ESM-2 150M improved luo_17 cold_target by $+0.069$ ($p<0.01$). BindingDB pre-training improved luo_17 cold_target by $+0.055$ ($p<0.01$) but caused negative transfer on yamanishi_e. Improvements are non-additive; C3 is the AUROC winner but at the cost of AUPRC degradation.

Objective 3 (Interpretability): Ensemble drug attribution achieves faithfulness advantage $+0.106$ (causal validity confirmed). Pharmacophore HBD precision at 10% is 3.8× random (biochemical relevance). DHFR binding-site AUROC=0.613, EF@10%=2.81× (structural correspondence without 3D supervision). Attribution is prediction-conditional – informative only when DTI confidence is high.

5.2 Limitations

- Yamanishi_e cold_target: near-random performance ($\leq$0.44 AUROC) for all models – dataset too small (26 proteins) for sequence-based cold-protein generalisation.
- Luo_17 cold_target anomaly: ConcatBaseline and GNN-Only outperform H-GNN-RNN-DTI – cross-attention may overfit to training protein embedding patterns.
- CPU training: limited epoch count and precluded ESM-2 650M evaluation and larger

BindingDB subsets.

- Interpretability scope: PDB validation limited to two representative pairs; cross-seed attribution consistency not systematically measured.

5.3 Future Work

1. GPU-accelerated training with ESM-2 650M for further cold_target improvement.
2. Adaptive/curriculum negative sampling that adjusts the ratio based on embedding model and dataset size.
3. Dataset-adaptive pre-training selection using domain alignment metrics (scaffold overlap, protein family similarity) to predict negative transfer before fine-tuning.
4. Cross-dataset transfer learning via domain-adaptive or meta-learning strategies.
5. Systematic faithfulness benchmark across multiple drug-protein pairs including cross-seed attribution consistency.
6. Integration of AlphaFold2-predicted 3D residue graphs for improved cold_target structural context.

5.4 Concluding Remarks

This research demonstrated that a carefully designed hybrid deep learning model – combining Hierarchical GAT, BiGRU, and Gated Cross-Attention with systematic enhancement strategies – achieves strong, reproducible performance across diverse DTI evaluation settings. The most significant finding is that training data distribution (negative-sampling ratio) has a larger impact on cold-start performance than architectural choices or embedding quality, suggesting that the primary bottleneck in DTI generalisation is often not the model architecture but the training distribution. The interpretability framework establishes that the model generates biologically meaningful explanations, recovering known binding pharmacophores and residues without structural supervision. Together, these contributions advance the field toward more reliable, interpretable, and generalisable computational tools for drug discovery.

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